

COMS30035, Machine learning: Unit Overview

James Cussens

School of Computer Science
University of Bristol

6th September 2026

Teaching Strategy

- ▶ In an ideal world this unit would both be:
 - Broad Covering many ML models and methods, and
 - Deep Enabling a thorough understanding of each of them
- ▶ But time constraints makes this impossible.
- ▶ The unit is reasonably broad but this means that some depth has been sacrificed.
- ▶ So there is less maths than there might be (sorry about that).
- ▶ A key objective is that when presented with a new data analysis task you can choose an appropriate method (since you know what modelling assumptions are made by that method), and moreover you know where to find and use the software that implements it.
- ▶ Another objective is that you appreciate the various statistical and computational trade-offs in the various approaches to machine learning.

Important topics that are missing

Pretty much all of the *Foundations* chapter of Murphy's *Probabilistic Machine Learning An Introduction* is missing . . .

- ▶ Lots of important maths (esp. linear algebra)
- ▶ Most of statistics
- ▶ Dimensionality reduction
- ▶ Optimisation algorithms
- ▶ Conjugate Bayesian analysis
- ▶ Reinforcement learning
- ▶ Inductive Logic Programming
- ▶ Causal machine learning (that's my area, so it's important of course)

Relation to other units

- ▶ Everyone on this unit will have done:
 1. *Mathematics for Computer Science A* (COMS10014)
 2. *Mathematics for Computer Science B* (COMS10013)
 3. *Algorithms and Data* (COMS20017)
- ▶ So we don't reteach material from those units (so e.g. no PCA).
- ▶ *Applied Data Science* (COMS30050) has a different 'angle': focusing on practical issues such as data cleaning, deploying ML solutions, ethics of ML, etc.
- ▶ And in parallel to this unit there is *Deep Learning* (COMS30177) which "introduces the students to the latest deep architectures," and allows students "to learn about the state-of-the-art in Deep Learning."
- ▶ In this unit we do one week on deep learning / neural networks.

Schedule

- Week 01 Intro and recap
- Week 02 Neural networks
- Week 03 Nonparametric methods
- Week 04 Bayesian methods
- Week 05 Clustering
- Week 07 Sequential data
- Week 08 Ensembles